



Streamlined strategy for discovering active compounds for nephrolithiasis treatment from herbal medicines using the fruit fly model

Yunuo Ren^{a,1}, Wen Zhang^{a,1}, Youxin Li^{a,1}, Dekun Wang^{b,1}, Weiwei Tang^{c,1}, Lingfei Wang^d, Xinghua Jin^a, Donghai Xia^{e,f}, Wei Yu^e, Yaodong Miao^g, Yingjie Ju^a, Shushen Sun^h, Tianxiang Liⁱ, Shaoshan Du^{h,*}, Zhimou Guo^{e,**}, Xiaoyue Tan^{b,***}, Zhiguang Yuchi^{a,****}, Yiwen Wang^{a,*****}

^a School of Pharmaceutical Science and Technology, Faculty of Medicine, Tianjin University, Tianjin, China

^b School of Medicine, Nankai University, Tianjin, 300071, China

^c School of Chemical Engineering and Technology, State Key Laboratory of Chemical Engineering, The Co-Innovation Center of Chemistry and Chemical Engineering of Tianjin, Tianjin University, Tianjin, 300072, China

^d First Teaching Hospital of Tianjin University of Traditional Chinese Medicine, 300193, Tianjin, China

^e Key Laboratory of Separation Sciences for Analytical Chemistry, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian, 116023, China

^f University of Chinese Academy of Sciences, Beijing, 100049, China

^g Second Affiliated Hospital of Tianjin University of Traditional Chinese Medicine, Tianjin, 300250, China

^h Department of Gastroenterology, Tianjin Jinnan Hospital (Tianjin University Jinnan Hospital), China

ⁱ Tianjin University of Traditional Chinese Medicine, Tianjin, China

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ABSTRACT

Ethnopharmacological relevance: Traditional Chinese medicines (TCMs) show promise for treating nephrolithiasis (kidney stones) with fewer adverse effects. However, identifying the key active compounds and understanding the therapeutic mechanisms of these herbs is challenging due to their complex compositions. Utilizing appropriate animal models is essential for overcoming this challenge and investigating the underlying mechanisms. **Aim of the study:** This study aims to establish a rapid discovery strategy to identify novel active components from TCMs using a *Drosophila melanogaster* (fruit fly) model.

Materials and methods: A discovery strategy for novel active components from candidate TCMs for the treatment of kidney stones was established using a fruit fly kidney stone model. The model was developed by feeding flies sodium oxalate (NaOx) to induce crystallization in their Malpighian tubules (MTs). Water extracts from various TCMs were screened for their efficacy against kidney stones. Effective extracts were then fractionated using solid-phase extraction and re-evaluated with the fruit fly model. These effective fractions were further analyzed using LC-MS and NMR to identify their compositions, and the identified compounds underwent additional testing with the fruit fly model to pinpoint the core active compounds. DHE staining, RT-qPCR, and *in vitro* crystallization assays were employed to explore the therapeutic mechanisms of both the effective TCM extracts and their core compounds, ensuring consistency in their mechanisms within the fly model. Finally, the efficacy of the core compounds was validated in a glyoxalate-induced mouse model.

Abbreviations: TCMs, Traditional Chinese medicines; MTs, Malpighian tubules; HH, water extract of Honghua; DS, water extract of Danshen; HQ, water extract of Huangqi; HYA, Hydroxysafflower yellow A; SAA, Salvanic acid A; CG, Calycosin-7-O-β-D-glucoside; PC, potassium citrate; PWEF, pure water elution fraction; EEf, ethanol elution fraction; ROS, reactive oxygen species; NAC, N-acetylcysteine; BUN, blood urea nitrogen; CaOx, calcium oxalate; NaOx, sodium oxalate; A_{sp}, aspect ratio; CRE, concentrations of creatinine; Gly, glyoxylic acid; DRE, drug extract ratio.

* Corresponding author.

** Corresponding author.

*** Corresponding author.

**** Corresponding author.

***** Corresponding author.

E-mail addresses: guozhimou@dicp.ac.cn (Z. Guo), xiaoyuetan@nankai.edu.cn (X. Tan), Yuchi@tju.edu.cn (Z. Yuchi), yiwen.wang@tju.edu.cn (Y. Wang).

¹ These authors contributed equally to this work.

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Results: In this study, a calcium oxalate kidney stone model was established using fruit fly. Three TCMs—Honghua (HH), Danshen (DS), and Huangqi (HQ)—were identified for their therapeutic effects against kidney stones from 15 candidates using the fruit fly model. The core compounds discovered were Hydroxysafflower yellow A (HYA) from HH, Salviatic acid A (SAA) from DS, and Calycosin-7-O- β -D-glucoside (CG) from HQ. HYA, SAA, and CG all demonstrated efficacy by inhibiting crystallization and reducing oxidative stress. Additionally, CG was also found to increase diuresis. The therapeutic effects of HYA, SAA, and CG were further validated in the mouse model.

Conclusion: We developed an efficient strategy for identifying core components of TCMs for kidney stones treatment using a fruit fly model combined with chemical isolation and identification techniques. This approach led to the discovery of three novel potential compounds for controlling kidney stones. Validated in mammalian models, these findings underscore the fruit fly model's utility in uncovering bioactive substances from TCMs, laying the groundwork for future research and the development of novel kidney stones therapies.

1. Introduction

Nephrolithiasis (kidney stones) is increasingly prevalent globally (Chewcharat and Curhan, 2021). Calcium oxalate (CaOx) stones account for 76% of cases (Singh et al., 2015). The high prevalence of kidney stones is associated with many factors such as high salt, high meat protein diet, and low water intake. In addition to this, family history, and systemic and urologic diseases are also associated with the development of kidney stones (Stamatelou and Goldfarb, 2023). The complexity of the multiple causes of kidney stones has not yet been fully resolved and has limited the development of effective therapeutic agents. Currently, the primary treatment for kidney stones involves minimally invasive kidney stone surgery, often complemented by medication (Fan et al., 2020). However, surgical interventions carry risks such as ureteral stenosis, renal injury, and urinary fistula. Moreover, the recurrence rate exceeds 50% within ten years (Tasian et al., 2017). Therefore, medication-based treatments are valuable, particularly post-surgery, to reduce recurrence and potentially avoid further operations. Various drugs are used to manage kidney stones through different mechanisms, like stone removal, dissolving, antispasmodic and analgesic effects, diuretic action, and anti-infective properties (Nirumand et al., 2018). However, existing drugs target specific symptoms but often come with side effects, and fail to provide a complete cure. This highlights the urgent need for new, effective drugs with fewer adverse effects for kidney stones.

Traditional Chinese medicine (TCM) has demonstrated efficacy in treating kidney stones (Siener, 2021). The diversity, minimal side effects, and high drugability of natural products make them valuable resources for developing novel and effective drugs for kidney stones. However, understanding the therapeutic mechanism of these herbs is challenging due to their complex compositions. The utilization of suitable animal models can play a pivotal role in surmounting this challenge, thereby facilitating the identification of the active compounds and the in-depth exploration of the underlying mechanisms.

Drosophila melanogaster (fruit fly) has been a cornerstone in genetic and medical research for over a century. Sharing approximately 75% of human disease-related genes, fruit flies exhibit significant genetic similarity to humans, making them an invaluable tool for studying human diseases and evaluating novel treatments (Graham and Pick, 2017). The Malpighian tubules (MTs) of fruit fly functionally resemble the human kidney and can develop CaOx crystals when exposed to lithogens such as sodium oxalate (NaOx) or ethylene glycol. Due to its simplicity and practicality, the fruit fly kidney stone model has become a widely adopted tool for exploring and validating the efficacy of drugs, including herbal medicines and compounds derived from TCMs (Chen et al., 2011; Wu et al., 2014). While this model has identified many effective herbal medicines for kidney stones, these herbs often consist of complex ingredient mixtures. Determining the core active compounds responsible for their therapeutic effects remains a crucial challenge, as identifying these compounds is essential for understanding the mechanisms and developing more effective herbal-based drugs. This step is pivotal in

harnessing the potential of TCM resources for the treatment of kidney stones.

In this study, we employed the fruit fly kidney stone model combined with a phytochemical work-up to establish a strategy for the rapid discovery and evaluation of core active components from candidate TCMs for treating kidney stones (Fig. 1). We have used this strategy to analyze active compounds for kidney stones treatment from TCMs.

2. Materials and Methods

2.1. Preparation for TCM extractions

TCMs were obtained from Tianjin University of Traditional Chinese Medicine. To prepare the TCM water extracts, 100 g of the herb was refluxed three times with 1000 mL of distilled water using a rotary evaporator, with each reflux lasting 2 h and the electric heating sleeve set to 50 V. The combined extracts were filtered through gauze and concentrated in a beaker until the volume was reduced to 100 mL, resulting in a 1 g/mL DRE stock solution.

2.2. Establishment of fly kidney stone model and treatment mediums

The normal diet (control group) consisted of 2.17 g yeast, 1.31 g sucrose, 1 g agar, 6.67 g corn syrup, 60 μ L propionic acid, 100 μ L methyl 4-hydroxybenzoate, and 100 mL distilled water. The lithogenic diet (NaOx group) was prepared in the same manner, with the addition of 0.1% NaOx. The treatment media were prepared by incorporating the respective drugs into the NaOx diet (Suppl. Table 1).

2.3. The observation and quantification of stone formation in the Malpighian tubules (MTs) of fruit fly

Wild-type female fruit flies aged 1–2 days were fed with the control medium, NaOx medium, or treatment medium for seven days. Twenty flies from each group were euthanized by CO₂ narcosis, and their MTs were carefully dissected under a stereomicroscope. The dissected MTs were transferred to glass slides with a drop of PBS buffer and examined under both normal and polarized light using a B302 biological microscope (Chongqing Optec Instrument Co., Ltd, China) at 40 $\times$ and 400 $\times$ magnification to identify and visualize CaOx crystals. A semi-quantitative method (Chen et al., 2011; Zhang et al., 2023) was used to score the crystal formation in the MTs, categorized as no crystals, minor (only fragmentary crystals in MTs), medium (crystal length more than half of the lumen diameter and fewer than ten stones per MT), and severe (crystal length more than half of the lumen diameter and more than ten stones per MT). Although this is a semi-quantitative approach, it takes into account both the number and size of crystals, providing a comprehensive assessment. To ensure consistency and minimize inter-observer variability, all scoring was performed under blinded conditions by trained personnel. Moreover, only compounds showing clear therapeutic effects in this initial evaluation are selected for further

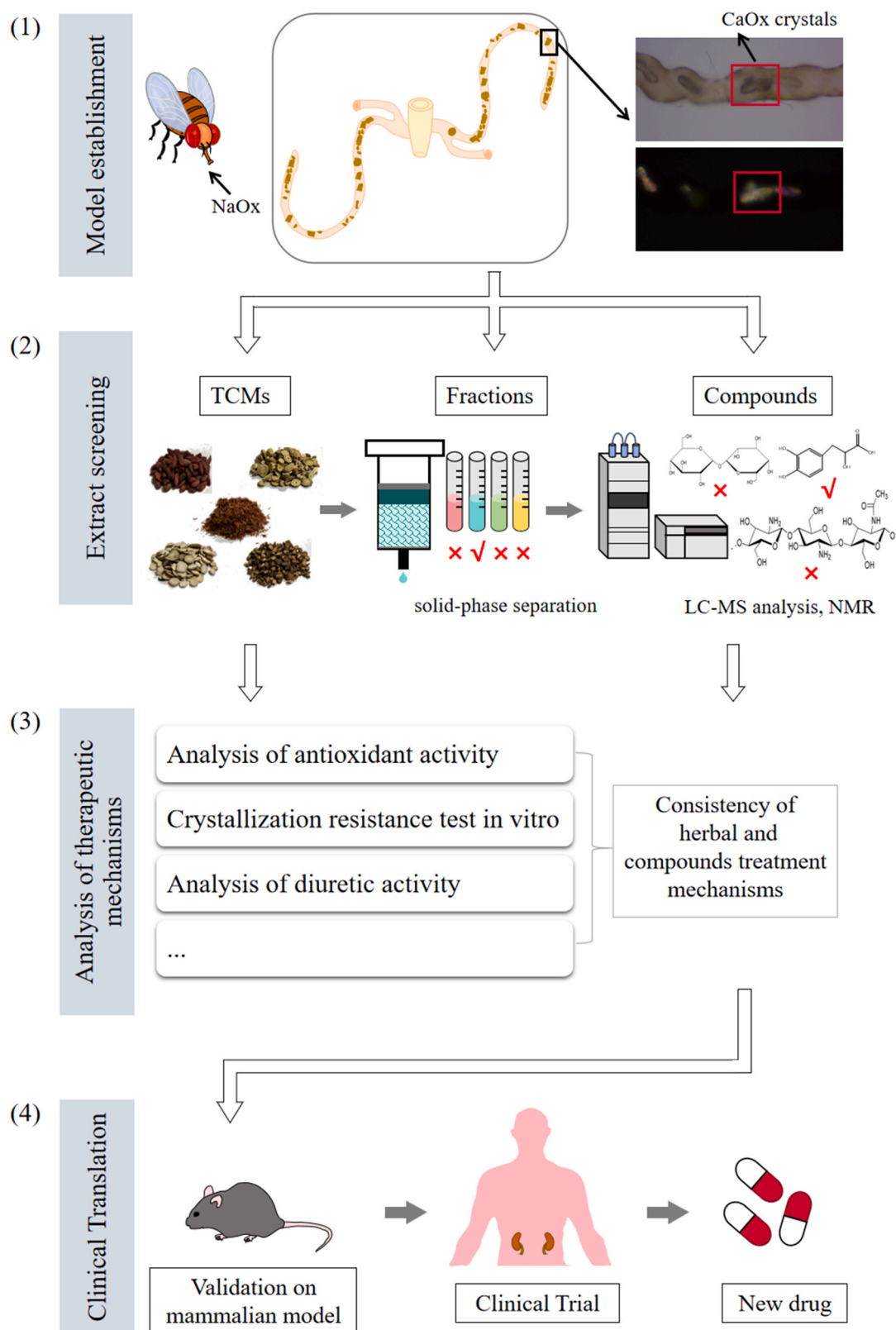


Fig. 1. The strategy for the rapid discovery of novel active components from candidate TCMs to treat kidney stones using the fly kidney stone model. (1) Model Establishment: The fruit fly kidney stone model is created by feeding flies with sodium oxalate and observing crystallization in their Malpighian tubules (MTs). (2) Extract screening: Various water extracts from TCMs are tested for their efficacy in treating kidney stones. The effective extracts are fractionated using solid-phase extraction. These fractions are then re-evaluated using the fruit fly model to identify the effective fractions. LC-MS analysis and NMR are employed to identify the compounds present in the active fractions. Subsequently, the fruit fly model is used to determine which of these compounds are the true core active substances with anti-kidney stone activity. (3) Analysis of therapeutic mechanisms: The therapeutic mechanisms of effective TCM extracts and their core compounds are analyzed using the fly model, comparing whether their mechanisms are consistent. (4) Clinical translation: The efficacy of the core compounds is validated in a mouse model, aligning with results from the fly models. Clinical trials will be conducted to complete the drug development process.

investigation, ensuring the robustness and reliability of the results.

2.4. Water extract separation and components identification

One kilogram of HH without ovary was refluxed in 10 L of water for 2 h, twice. One kilogram of DS and HQ was refluxed in a 60% ethanol/40% water mixture (10 L) for 2 h, twice. All three extracts were concentrated to 4 L using a rotary evaporator under a high vacuum at 60 °C. If present, ethanol or formic acid should be removed simultaneously. Each concentrated extract solution was filtered through a ceramic microfiltration membrane, and the permeate was collected for further preparative liquid chromatography (Prep-LC) separation. In the Prep-LC step, an RP-C18 (250 × 30 mm i. d.) column was used to prepare the herbal fractions. The flow rate was 20 mL/min, and the loading amount was 2.5% (based on the dry weight of the extract). Afterward, all fractions were rotary evaporated and freeze-dried. The effective target fraction underwent analysis via LC-MS (Suppl. Fig. 2). Based on their molecular weights, the main peaks were precisely identified to predict the potential major compounds. Standard compounds HYA (PS0782 - 0500, Chengdu Push Bio-Technology Co., Ltd., China), SAA (S31458 - 1g, Shanghai Yuanye Bio-Technology Co., Ltd., China), and CG (PS0409 - 0500, Chengdu Push Bio-Technology Co., Ltd., China) were purchased to ensure the purity of the samples for subsequent *in vivo* analyses of single compounds. The structures of these standard compounds were rigorously verified using NMR (Suppl. Fig. 3). These standard products were used in subsequent experiments on fruit flies and mice.

2.5. Bulk crystallization *in vitro*

The bulk crystallization assay was conducted following the protocol established by Li et al. (Li et al., 2020; Li et al., 2022). *In vitro* crystallization tests were performed at room temperature in 25 mL beakers, each with a clean coverslip placed at the bottom to gather crystals for additional analysis. The TCM extract stock solutions were filtered through 0.22 µm filters. Solutions containing 150 mM NaCl, 0.56 mM CaCl₂, 0.56 mM Na₂C₂O₄, and x mM/x mg/mL inhibitors were placed on a heating plate at 37 °C for 3 days in a static state. The range of x is 0.03–0.67 mM for CA, 0.02–0.4 mg/mL DRE for HH, DA and HQ, 0.0065–0.52 mM (0.004–0.32 mg/mL) for HYA, 0.021–1.61 mM (0.004–0.32 mg/mL) for SAA, or 0.004–0.2 mg/mL for HQ-30% EEF. After this period, the coverslips were gently removed from the growth solution, and the morphology of the crystals was characterized using a transreflective polarizing microscope (Beijing Precise Instrument Co., Ltd, China), specifically, the aspect ratio (A_{sp}) of the crystals was analyzed. Three replicates are carried out for each experiment. The morphology of crystals with whiskers was characterized by a scanning electron microscope (SEM, Hitachi, TM3000).

2.6. Establishment of mouse kidney stone model

Male C57BL/6J mice, aged 6–8 weeks and weighing between 18 and 22 g, were procured from Beijing Sinogenetic Biotechnology Company (Beijing, China). The mice were systematically randomized into six experimental groups including control, glyoxylate (Gly), positive control (citric acid), and three treatment groups (HYA, CG, and SAA). Kidney stones were induced by administering Gly intraperitoneally at a dose of 75 mg/kg body weight for six consecutive days. For the treatment groups, doses of 40 mg/kg, 30 mg/kg, and 25 mg/kg body weight, respectively, were administered intraperitoneally, with a 10-h interval after Gly administration, for six consecutive days. Control groups received equivalent volumes of normal saline or 50 mg/kg body weight of citric acid via intraperitoneal injection, serving as negative and positive controls, respectively. On the 7th day, mice were euthanized via cervical dislocation under deep anesthesia (5% isoflurane), and serum and kidney tissues were collected for subsequent analyses.

Renal function was evaluated by measuring serum creatinine (Scr)

and blood urea nitrogen (BUN). Renal injury was assessed using Periodic acid-Schiff (PAS) staining, while crystal deposition was examined via Hematoxylin-eosin (HE) staining under polarized light microscopy.

All mice experiments were performed by ARRIVE guidelines (Percie du Sert et al., 2020) and were approved by the Institutional Animal Care and Use Committee of Nankai University (the approval number and date: 2023-SYDWLL-000185; February 2nd, 2023).

2.7. Hematoxylin-Eosin staining and polarized light microscopy analysis

Paraffin-embedded kidney tissue sections (5-µm thick) were deparaffinized in xylene and rehydrated through a graded ethanol series. Following dehydration, sections were stained with hematoxylin and eosin (H&E) and mounted for microscopic examination. Crystal deposition was quantified using polarized light microscopy (Zeiss Primotech, Oberkochen, Germany).

The detailed method information is provided in the Materials and Methods section of the Supporting Information.

3. Results

3.1. Screening traditional Chinese medicines with therapeutic effects on kidney stones

To assess the therapeutic efficacy of various TCMs on kidney stones, we established a fruit fly kidney stone model by administering NaOx to the flies for seven days (Chen et al., 2011). In the control group, only 5% of fruit flies exhibited minor CaOx crystal formation in their MTs, with the majority showing no stone formation. In contrast, 90% of fruit flies in the NaOx-induced group displayed severe stone formation. Citrate salts (such as potassium citrate and sodium citrate) (Phillips et al., 2015) are the most commonly used drugs for the treatment of kidney stones and are also widely employed as positive controls in the majority of related studies. In this study, the positive control group treated with 5% potassium citrate (PC) exhibited a marked improvement, with 70% of the flies showing either no crystals or only minor crystal formation in their MTs (Fig. 2).

Based on the medicinal substances recorded in “Ben Cao Bei Yao”, we invited traditional Chinese medicine (TCM) experts to compile a list of approximately 100 herbs traditionally used to treat Shi Lin (the TCM term for kidney stones). From this list, we randomly selected 15 herbal medicines for efficacy evaluation. The TCMs tested were Baishao (*Cynanchum otophyllum* Schneid), Sangbaipi (*Morus alba* L.), Jixueteng (*Caulis Spatholobi*), Huzhang (*Polygonum cuspidatum* Sieb. et Zucc), Xuanshen (*Scrophularia ningpoensis* Hems), Shanzha (*Crataegus pinnatifida* Bunge), Yinchen (*Artemisia capillaris* Thunb), Difuzi (*Kochia scoparia* (L.) Schrad), Yimucao (*Leonurus japonicus* Houtt), Gouteng (*Uncaria rhynchophylla*), Luoshiteng (*Trachelospermum jasminoides* (Lindl.) Lem), Xiaoji (*Herba Cirsii Setosi*), Danshen (*Salvia miltiorrhiza* Bunge), Honghua (*Carthamus tinctorius* L.), and Huangqi (*Astragalus membranaceus* (Fisch.) Bunge). We chose water extraction in our study to accurately replicate traditional Chinese medicine decoctions, ensuring maximal preservation and effective exertion of their medicinal value.

During the initial screening phase, we standardized the dosage of TCM extract to 0.2 g/mL drug extract ratio (DRE) (Zhang et al., 2023). All the selected TCMs demonstrated varying degrees of effectiveness. Notably, Honghua (HH) and Huangqi (HQ) demonstrated robust therapeutic effects, with 100% of flies showing no or only minor CaOx crystal formation. Most other herbs showed partial effects, with fewer than 50% of flies exhibiting no or minor stone formation. Danshen (DS) demonstrated moderate effectiveness at 0.2 g/mL DRE, with 75% of flies showing no or minor stones in MTs (Fig. 2). Its effectiveness increased to 100% at the optimal dosage of 1 g/mL DRE (Fig. 3-B). We next determined that the optimal dosage for HH and HQ were 0.2 g/mL DRE (Fig. 3-A) and 0.5 g/mL DRE (Fig. 3-C), respectively, both achieving 100% efficacy in the fruit fly model. These optimal dosages were

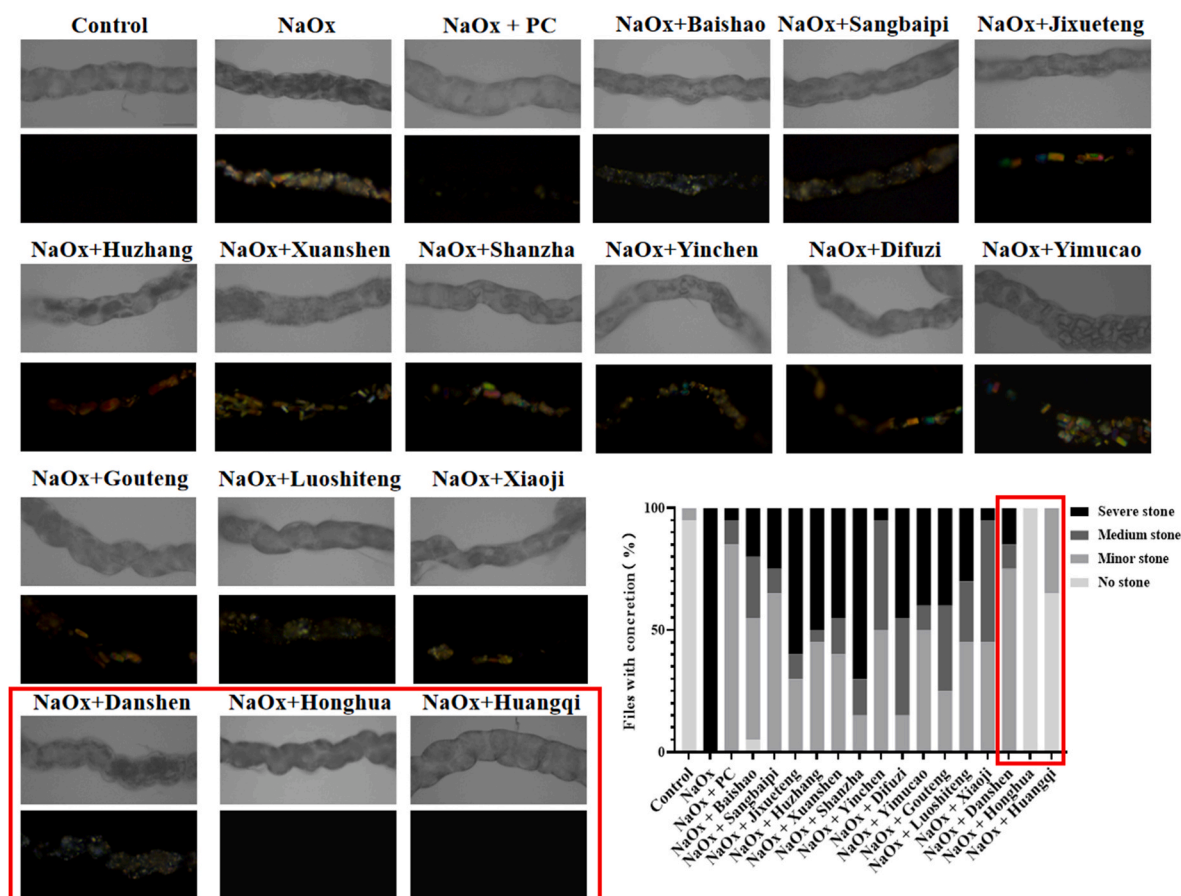


Fig. 2. Effects of 15 TCMs on CaOx stone formation in the MTs of fruit flies treated with NaOx. Representative images of MTs in fruit fly, observed under normal light and polarized light microscopy. Flies were fed for seven days on standard medium, 0.1% NaOx fly medium, and 0.1% NaOx medium supplemented with 0.2 g/mL DRE water extract of 15 types of TCMs. The scale bar represents 50 μ m. The percentage histogram illustrates the varying degrees of CaOx crystal formation observed in the MTs of fruit flies ($n = 20$ for each group).

selected for subsequent experiments.

3.2. Exploring the core active compounds of HH, DS, and HQ

HH, DS, and HQ have all been reported to have therapeutic effects in treating chronic kidney diseases; however, their specific active substances remain unclear (Huang et al., 2023; Liu et al., 2019, 2023). To address this, we first employed solid-phase extraction to separate HH, DS, and HQ water extracts into distinct fractions, which were then individually evaluated for therapeutic efficacy.

HH was fractionated into four fractions: pure water elution fraction (HH-PWEF), 20% ethanol elution fraction (HH-20% EEF), 30% ethanol elution fraction (HH-30% EEF), and 50% ethanol elution fraction (HH-50% EEF). Among these, only HH-20% EEF demonstrated significant therapeutic effects at a concentration of 1 mg/mL, with 85% of flies showing no or minor CaOx crystal formation (Fig. 4-A3, Suppl. Fig. 2A, Suppl. Fig. 4).

DS was fractionated into three fractions: DS-PWEF, DS-20% EEF, and DS-40% EEF. Remarkably, DS-PWEF exhibited the strongest therapeutic activity, with a notable reduction in crystal formation at a concentration of 5 mg/mL, as 75% of fruit flies showed no or minor crystals. In contrast, DS-20% EEF and DS-40% EEF displayed only mild inhibitory effects on stone formation (Fig. 4-B3, Suppl. Fig. 2B, Suppl. Fig. 5).

HQ was fractionated into five fractions: HQ-PWEF, HQ-15% EEF, HQ-50% EEF, HQ-70% EEF, and HQ-30% EEF. Among these, only HQ-30% EEF demonstrated significant efficacy, with an 85% reduction observed at a concentration of 1 mg/mL (Fig. 4-C3, Suppl. Fig. 2C, Suppl. Fig. 6).

Using LC-MS, we identified a major compound in each of the active fractions: Hydroxysafflower yellow A (HYA) in HH-20% EEF, Salvianic acid A (SAA) in DS-PWEF, and Calycosin-7-O- β -D-glucoside (CG) in HQ-30% EEF (Suppl. Fig. 2). Subsequently, we purchased the standard compounds of HYA, SAA, and CG. To confirm their accuracy, we thoroughly validated the structures of these compounds using NMR (Suppl. Fig. 3). After this verification, we conducted experiments to test the antilithiasis activity of the standard compounds on fruit flies.

Evaluating the therapeutic effects of HYA, SAA, and CG revealed a significant reduction in NaOx-induced crystal formation in fruit flies when treated with each of these three compounds individually. At 1 mg/mL, each compound achieved a 70% reduction in stone formation, showing potent therapeutic effects (Fig. 4-A4, B4, C4, Suppl. Fig. 7). Although the therapeutic activities of HYA, SAA, and CG have been identified for various diseases (Bao et al., 2022; Tsai et al., 2019; Wang et al., 2020, 2021; Xu et al., 2022; Zhang et al., 2019), this study is the first to confirm their effectiveness in treating kidney stones using animal models.

3.3. The therapeutic mechanisms of HYA, SAA, and CG in kidney stones treatment

To explore the therapeutic mechanisms of HH, DS, HQ, and their active compounds, we investigated their effects on reactive oxygen species (ROS) levels in Malpighian tubules (MTs), crystal morphology *in vitro*, and urinary excretion levels in fruit flies. These tests aimed to assess the compounds' abilities to reduce oxidative stress, inhibit CaOx crystallization, and enhance urinary excretion—three key mechanisms

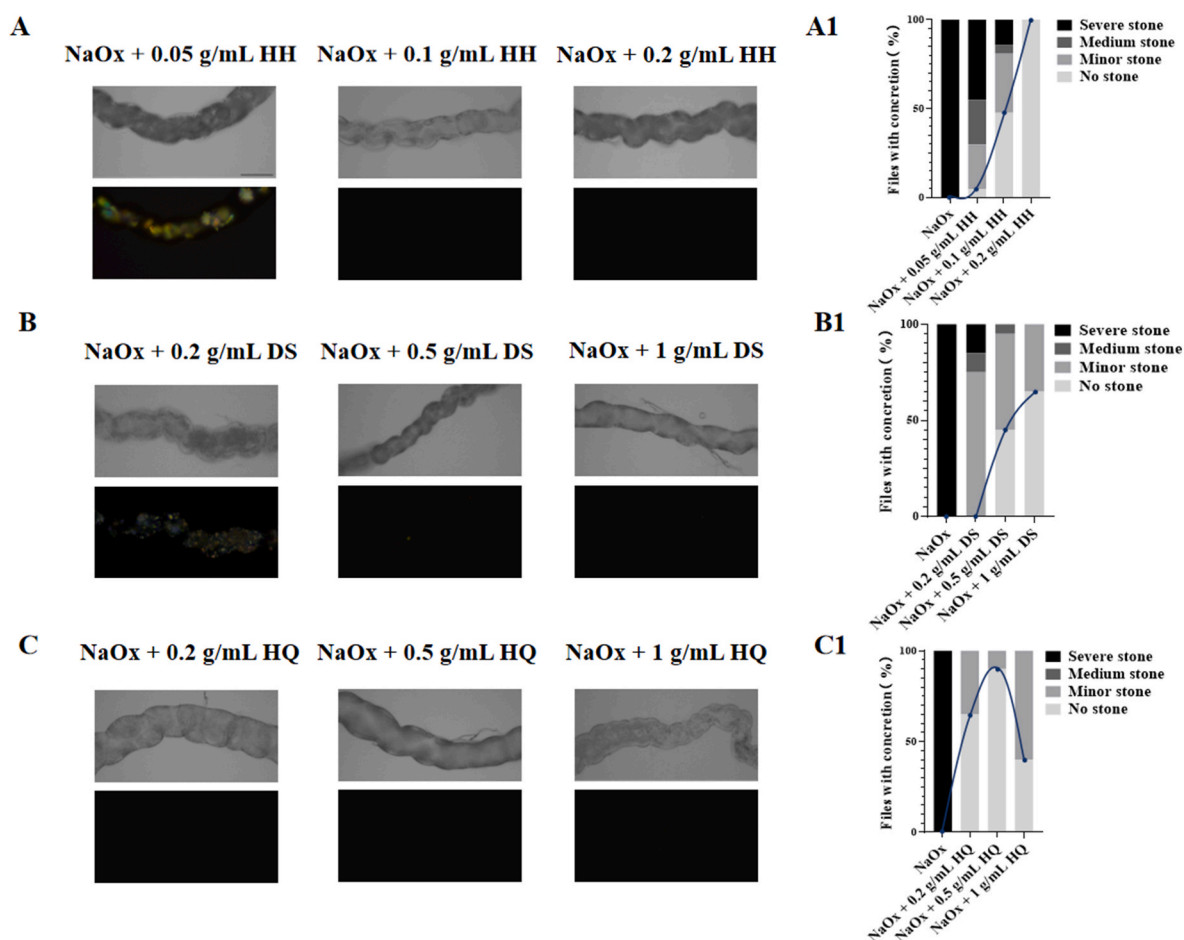


Fig. 3. Effect of different dosages of HH, DS, and HQ water extract on CaOx stone formation in the MTs. Representative images captured under normal light and polarized light microscopy show the MTs of fruit fly after seven days of feeding on fly mediums containing 0.1% NaOx and varying dosages of water extract of HH (A1), DS (B1), and HQ (C1) separately. The scale bar represents 50 μ m. The percentage histograms (A2, B2, and C2) illustrate the varying degrees of CaOx crystals observed in the MTs of fruit flies ($n = 20$ for each group) treated with different doses of HH, DS, and HQ.

known to affect kidney stone formation.

We stained MTs with DHE to assess the ROS levels. The NaOx-induced group exhibited significantly higher ROS levels compared to the control group. All treatment groups showed a significant decrease in ROS levels (Fig. 5-A, A1, Suppl. Fig. 8). Additionally, we analyzed the expression levels of genes encoding antioxidant enzymes, namely *Sod1*, *Sod2*, and *Cat*, which are positively associated with oxidative stress. The genes were significantly down-regulated in the treatment groups (Fig. 5-B). These findings suggested that HH, DS, and HQ, and their active components HYA, SAA, and CG may alleviate kidney stones by reducing the ROS production and oxidative stress in the MTs of fruit flies. It is generally believed that oxidative stress induced by oxalic acid toxicity leads to inflammation, cellular damage, and apoptosis in the urinary system, promoting crystal formation and deposition by releasing membranes and vesicles that serve as nucleation sites (Fasano and Khan, 2001; Khan, 2011). However, data from fruit flies fed a combination of N-acetylcysteine (NAC) and NaOx showed that merely inhibiting oxidative stress does not completely treat kidney stones (Suppl. Fig. 9). Thus, while reducing oxidative stress supports stone prevention, it is not the primary mechanism for treating kidney stones.

The primary cause of kidney stones is the excessive concentration of substances in urine that leads to the crystallization and growth of minerals, particularly CaOx (Li, S. et al., 2022). Potential strategies to prevent kidney stone formation include developing substances that either prevent crystallization or alter crystal morphology to reduce attachment to renal epithelial cells (Moe, 2006; Qaseem et al., 2014). For example,

citric acid (CA) inhibits CaOx crystal formation by binding to calcium, making it a commonly used drug for this purpose (Grohe et al., 2011). We also conducted an *in vitro* study to evaluate the effects of HH, DS, HQ, and their active components on crystal morphology which can be characterized by the [001]/[010] aspect ratio (A_{sp}) of the crystals (Li et al., 2020; Li et al., 2022) (Fig. 6-A).

SEM images of CaOx crystals (Fig. 6-A) reveal the apparent binding specificity of inhibitors. In the model group, where CaOx crystals were synthesized using 0.56 mM $\text{Na}_2\text{C}_2\text{O}_4$ to a CaCl_2 solution, the crystals appeared as regular hexagons after 3 days under static conditions. In the positive control group, which received 0.03 mM–0.67 mM of CA (Li et al., 2020; Li et al., 2022), a quadrilateral morphology was observed, indicating growth inhibition in both the [010] and [001] directions. Additionally, the A_{sp} of the crystals varied as a function of the inhibitor concentration (Fig. 6-B).

Treatment with 0.02–0.4 mg/mL DRE HH, DS, and HQ (0.02–0.4 mg/mL DRE) effectively inhibited the A_{sp} of the crystal and demonstrated concentration-dependent inhibition of the A_{sp} of the crystal. Moreover, when the inhibitors HH and DS were added at a dosage of 0.4 mg/ml DRE, the crystal morphology appeared nearly circular and the crystals of HQ were octagonal at the same concentration. For the core active compounds, HYA (0.0065–0.52 mM) and SAA (0.021–1.61 mM) were also successful in preventing crystal formation and showed concentration-dependent inhibition of the A_{sp} within the crystal. Due to CG's limited solubility, HQ-30% EEf, containing CG, was used for *in vitro* experiments. Similarly, HQ and HQ-30% EEf (containing CG) were

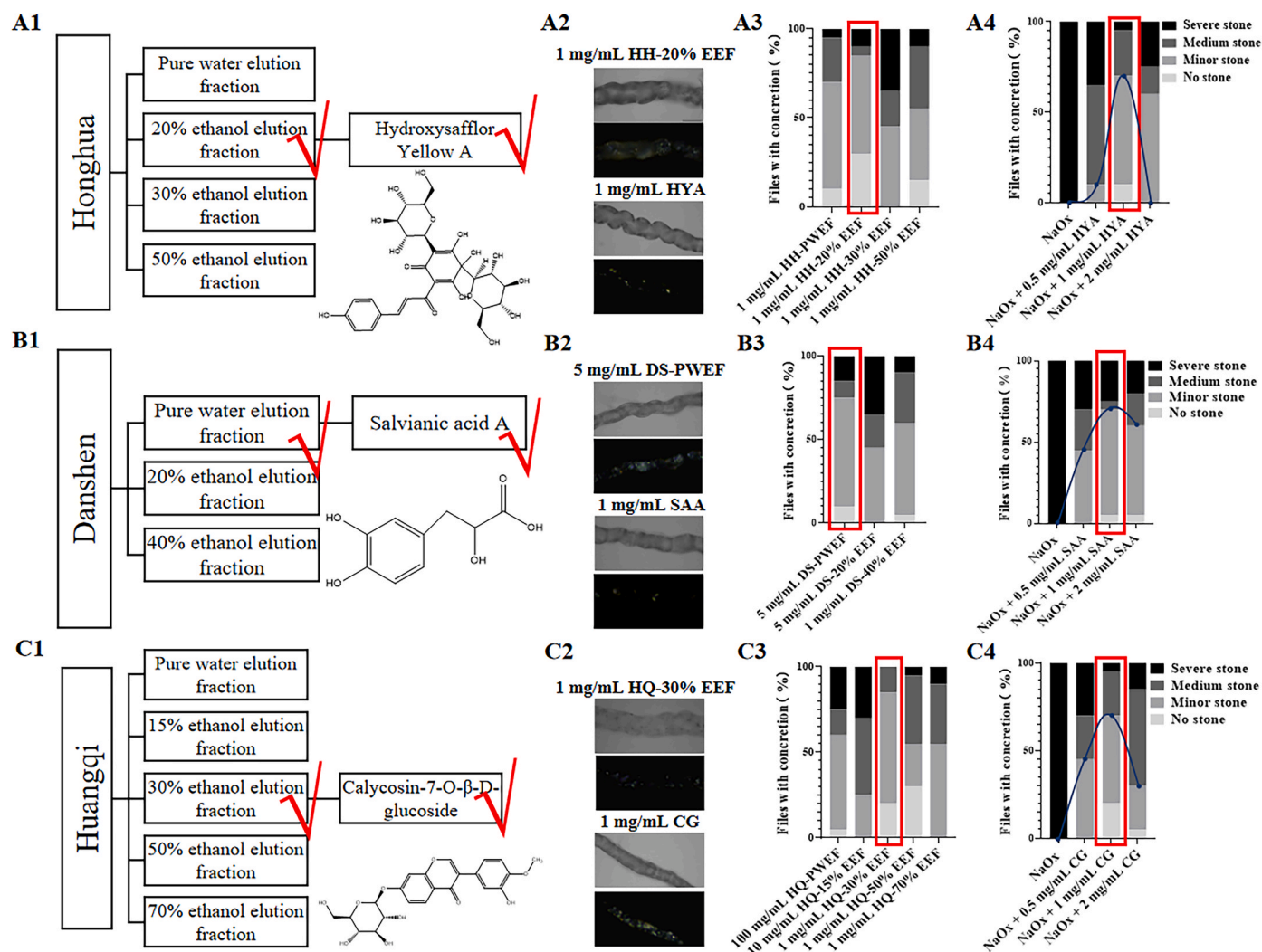


Fig. 4. Screening of active components in HH, DS, and HQ. The results of solid-phase extraction for water extracts of HH (A1), DS (B1), and HQ (C1), along with the structure of identified active compounds HYA, SAA, and CG, are presented. The ones marked in red are the effective fractions and active compounds. Representative images are shown under normal light and polarized light microscopy of MTs from fruit flies fed for seven days on media containing 0.1% NaOx and optimal concentrations of HH-20% EEf with HYA (A2), DS-PWEF with SAA (B2), and HQ-30% EEf with CG (C2). The scale bar represents 50 μ m. The percentage histograms depict varying levels of CaOx crystals in the MTs of fruit flies ($n = 20$ per group) treated with optimal concentrations of different fractions of HH (A3), DS (B3), and HQ (C3), as well as different doses of HYA (A4), SAA (B4), and CG (C4). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

effective in inhibiting crystal formation, showing concentration-dependent inhibition of the A_{sp} crystal within the range of 0.004–0.2 mg/mL. Additionally, HYA, SAA, and HQ-30% EEf demonstrate significantly superior performance, preventing growth at concentrations lower than that of CA.

Increasing urination is a key mechanism for treating kidney stones, as it reduces calcium and oxalate concentrations in urine and helps flush out small stones and fragments (Chen et al., 2020). Diuretics like furosemide and spironolactone, along with herbs such as *Haijinsha* (*Spora Lygodii*) and *Jinqiancao* (*Lysimachia christinae*), enhance urine production and ureter peristalsis, preventing stone formation (Gohel and Wong, 2006; Lyall et al., 2023). In this study, urine output was monitored using fruit fly medium containing blue dye. Flies exposed to NaOx exhibited significantly lower urine output compared to the control group. Among the six drugs investigated, only HQ exhibited a significant diuretic effect (Fig. 7-A). Additionally, the expression of genes associated with urination in fruit fly, including *Dh31*, *Nhe2*, *cac*, *Ca-α1D*, *Prestin*, *JYalpha*, and *Nha1*, were all up-regulated following HQ treatment (Fig. 7-C). In contrast, flies treated with CG (from HQ) had higher urine output compared to the NaOx-induced group but lower than the control group

(Fig. 7-A). Thus, CG does not significantly increase urine output for crystal removal. The increased urine output in the CG group is likely due to the clearance of kidney stones, rather than a direct diuretic effect. The observed diuretic activity in HQ is likely attributed to its other components.

In summary, the mechanisms of action of the three TCM water extracts generally align with their components, HYA, SAA, and CG. HH, DS, and HQ, along with their active compounds, primarily inhibit kidney stones formation by reducing oxidative stress and inhibiting CaOx crystallization through multi-target synergies. Additionally, HQ may also aid in treating renal calculi by increasing urinary excretion, likely due to other unidentified compounds. The alignment of mechanisms between the aqueous extracts and their principal core compounds confirms the identification of effective active compounds for treating kidney stones.

3.4. Treatment of kidney stones in the mouse model with HYA, SAA, and CG

To further evaluate the therapeutic potential of the three candidate

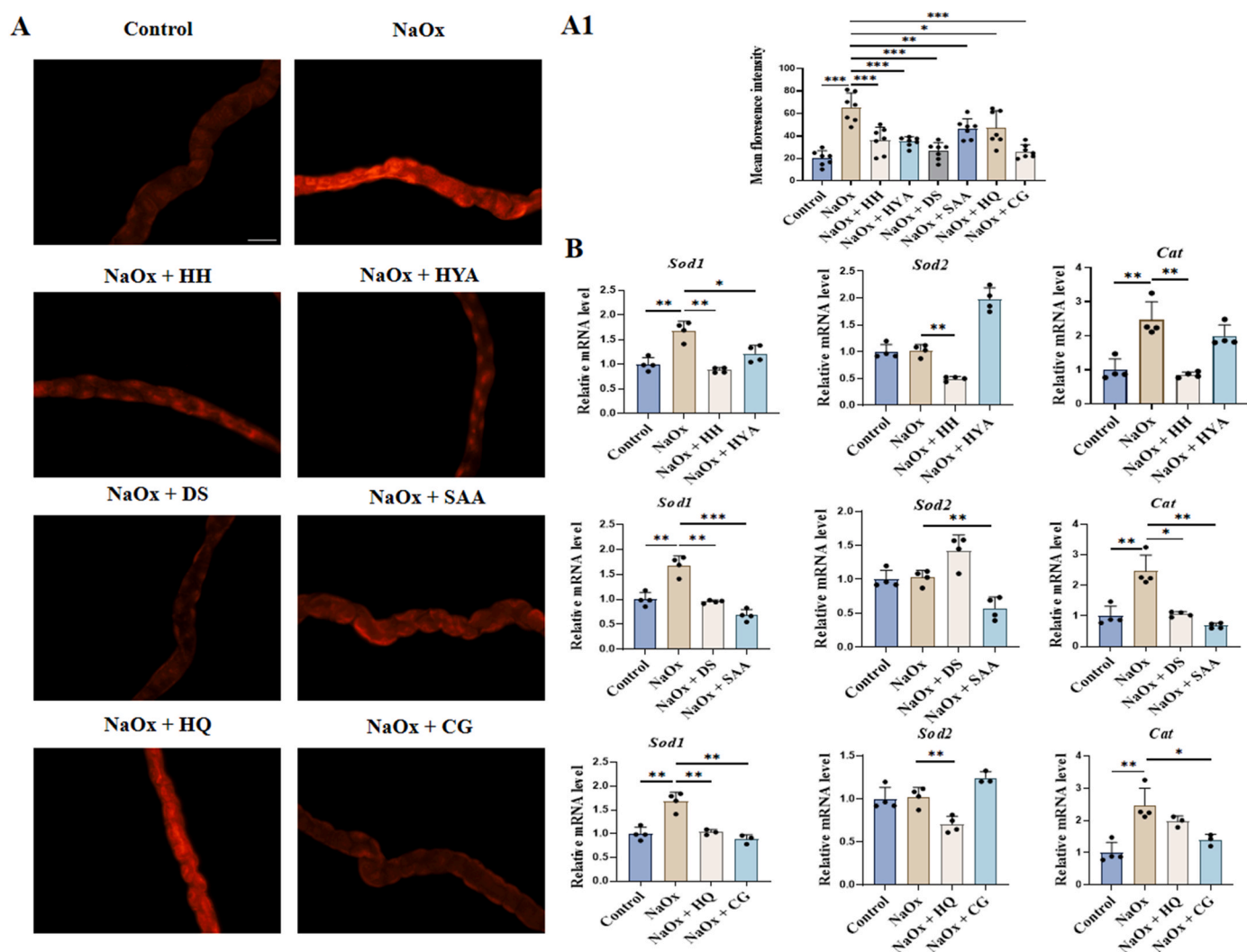


Fig. 5. HH, DS, HQ as well as their compounds HYA, SAA, and CG decreased the ROS levels of MTs. Representative DHE strain images of MTs (A) and the quantification of DHE staining (A1) for the fruit flies fed for seven days on fly mediums containing 0.1% NaOx and HH, DS, HQ, HYA, SAA, and CG separately are shown. Scale bar: 50 μ m. Data are represented as mean $\pm$ standard deviation (Mean $\pm$ SD) and analyzed using One-Way ANOVA (n/s, $p > 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$). The expression levels of key genes associated with oxidative stress were quantified (B), with ribosomal protein 49 (rp49) used as the reference gene. All other genes were normalized by *rp49*. Data were normalized to the values of the control group.

compounds against mammalian kidney stones, we established a CaOx kidney stone model in mice through six consecutive days of intraperitoneal glyoxylic acid (Gly) administration. Treatment compounds were administered via intraperitoneal injection 10 h post-Gly injection (Fig. 8-A). Serum creatinine and blood urea nitrogen (BUN) levels were significantly elevated in the Gly-mediated kidney stone mouse group (Gly group) compared to the control group, indicating substantial renal impairment. Citric acid (positive control) treatment effectively normalized these parameters to near-control levels (Fig. 8-B, C). Additionally, tubular injury was characterized by tubular epithelial necrosis, brush margin loss, tubular formation, and tubular dilatation. The tubular injury index was markedly higher than in the Gly group versus controls, while the citric acid group showed intermediate improvement (Fig. 8-D1, D2). Polarized light microscopic examination of H&E-stained kidney sections revealed distinct patterns of CaOx crystal deposition across experimental groups. control animals showed complete absence of crystals, Gly-treated mice exhibited extensive crystal deposition throughout renal tissues, while the citric acid-treated group demonstrated a significant reduction in crystal burden compared to the Gly group (Fig. 8-E1, E2).

All three treatment groups (HYA, SAA, CG) showed significant

improvement across all renal parameters versus the Gly group. Serum creatinine and BUN levels were significantly decreased in all three treatment groups compared to the Gly group, and the levels were close to those of the positive control group (Fig. 8-B, C). The tubular injury indexes of the three treatment groups were also significantly reduced to the Gly group; HYA, SAA, and CG all exhibited inhibitory effects on stone formation (Fig. 8-D, E). Notably, SAA group exhibited the most pronounced therapeutic effect, with outcomes closely approaching those of the positive control, though full recovery to baseline levels was not achieved. In summary, HYA, SAA, and CG can inhibit stone formation in mice, consistent with the results of fly experiments.

Through systematic screening of 15 TCM candidates, we identified three promising anti-nephrolithic herbs: HH, DS, and HQ. We successfully identified the core active compounds responsible for these herbs' anti-kidney stone effects. These identified compounds exhibited potent antioxidant activity and effectively inhibited the crystallization process. Finally, the results were validated in a mouse model. These findings not only validate the translational relevance of our fruit fly screening platform for mammalian kidney stones research, but also highlight the potential clinical applications of these natural compounds in kidney stone treatment.

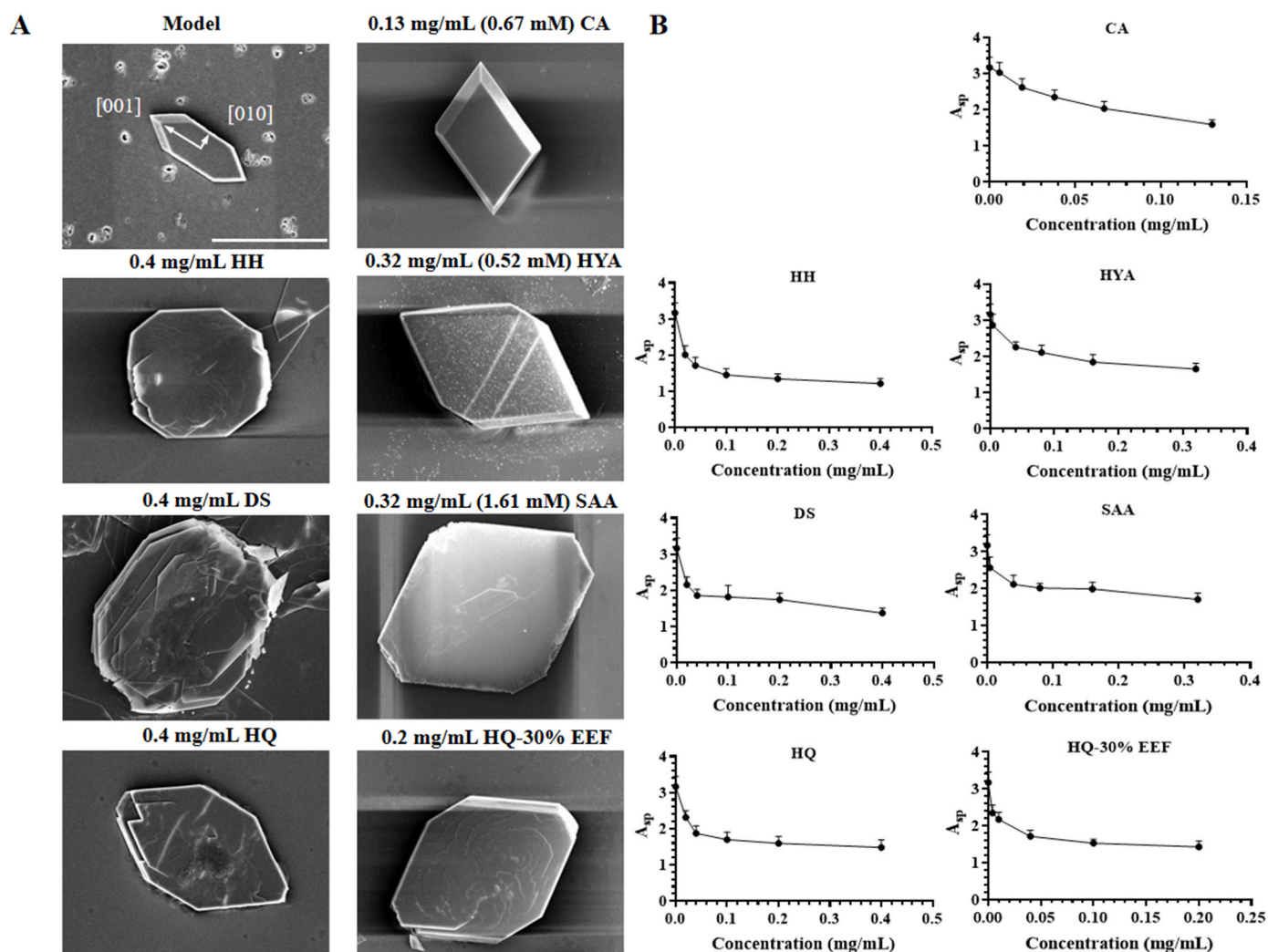


Fig. 6. HH, DS, HQ, and their compounds HYA, SAA, and CG inhibited the crystal morphology *in vitro*. SEM images of crystals in the absence of inhibitors and the presence (A). Inhibition percentage of CaOx crystallization as a function of inhibitor concentration (B). Data are represented as mean $\pm$ standard deviation (Mean $\pm$ SD). Scale bar: 10 μ m.

4. Discussion

In this study, we developed a strategy integrating the fruit fly model with phytochemical separation techniques to efficiently screen and identify their active compounds from TCM for treating calcium oxalate kidney stones. We use a fruit fly model induced with sodium oxalate NaOx to mimic kidney stone formation (Chen et al., 2011). Our screening identified three effective herbs—Honghua (HH), Danshen (DS), and Huangqi (HQ)—from 15 TCM candidates. Based on Chinese medicine theories, HH and DS are known to enhance blood circulation and alleviate blood stasis, while HQ is believed to replenish qi and boost resistance, all contributing to kidney stone treatment. Huangqi and Danshen Decoction are also established for chronic kidney disease, though their specific active substances remain unclear (Huang et al., 2023; Liu et al., 2019, 2023). In addition, these three herbs are also included in the ancient prescriptions for the treatment of kidney stones (Table 1).

To identify the primary active compounds of these three herbs for treating kidney stones, we integrated the fly model with chemical isolation and identification techniques, including solid-phase extraction, LC-MS, and NMR. We identified HYA in HH, SAA in DS, and CG in HQ. These compounds achieved a 70% reduction in stone formation in the fly model, suggesting they are likely the core components responsible for the therapeutic effects of these herbs. Both the three TCMs and these

three active compounds have not demonstrated significant toxicity or adverse effects in existing clinical and cellular studies (Jiao et al., 2025; Li et al., 2009; Liu et al., 2025; Xu et al., 2024; Yi et al., 2022; Yu et al., 2025), suggesting a wide safety margin for their use.

HYA has demonstrated potential therapeutic effects for myocardial ischemia (Bao et al., 2022), osteoporosis (Wang et al., 2021), and other conditions. SAA is promising for cardiovascular diseases (e.g., myocardial ischemia and reperfusion, atherosclerosis, hypertension), and cerebral pathologies and diseases (e.g., ischemia, cognitive decline, anxiety) (Wang et al., 2020; Zhang et al., 2019). CG has been identified as a potential treatment for hepatic steatosis (Xu et al., 2022) and myocardial injury (Tsai et al., 2019). However, this study is the first to confirm that HYA, SAA, and CG are effective in treating kidney stones using animal models. The established mechanisms for treating kidney stones involve inhibiting crystal formation, increasing urinary excretion levels, and decreasing oxidative stress.

The primary cause of kidney stones is the excessive concentration of substances in urine that leads to the crystallization and growth of minerals, particularly CaOx (Li, S. et al., 2022). Potential strategies to prevent kidney stones formation include developing substances that either prevent crystallization or alter crystal morphology to reduce attachment to renal epithelial cells (Moe, 2006; Qaseem et al., 2014). For example, Citric acid inhibits calcium oxalate crystal formation by binding to calcium, making it a commonly used drug for this purpose (Grohe et al.,

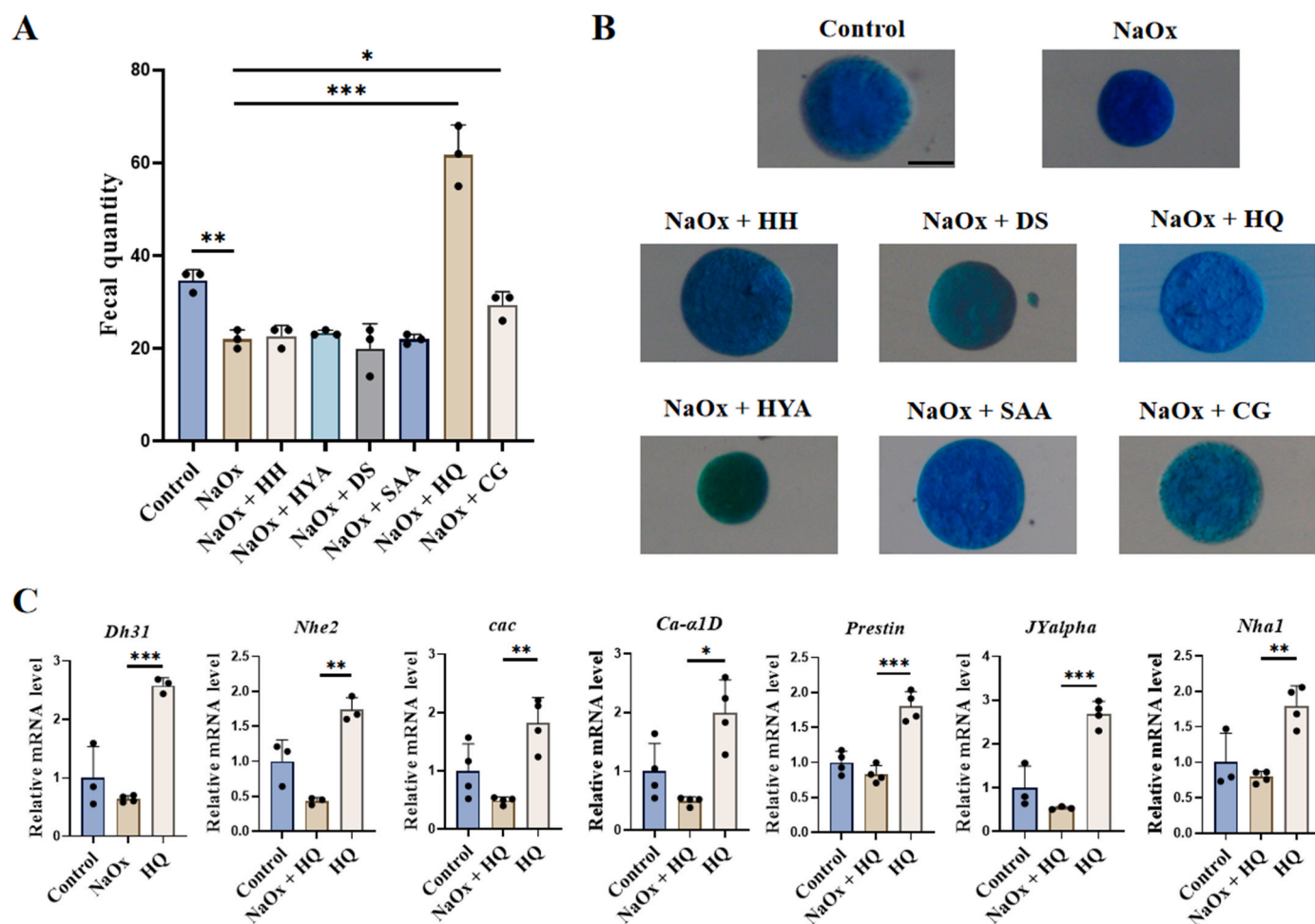


Fig. 7. HQ increased fruit flies' excretion level. The quantities excreted by different groups of fruit flies ($n = 60$ for each group) (A) and images of fruit fly deposits (urine and feces). Scale bar: 200 μm . (B) are displayed. Data are presented as mean $\pm$ standard deviation (Mean $\pm$ SD), and analyzed by One-Way ANOVA (n/s , $p > 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$). Additionally, the expression of key genes associated with urination is analyzed. (C). The reference gene was *rp49*, and all other genes were normalized by *rp49*. Data were normalized to the values of the control group.

2011). A method established by Li et al. (Li et al., 2020; Li et al., 2022) for the stable screening of the effects of inhibitors on crystals was adopted in our analysis. We utilized this method to examine the impact of three herbs and their active compounds on crystalline morphology. Our findings indicate that all three compounds effectively reduce the A_{sp} of the crystals, demonstrating a concentration-dependent effect.

Oxalic acid toxicity induces oxidative stress in the urinary system, leading to inflammation, cellular damage, and apoptosis, which can harm urinary epithelial cells. The cellular degradation releases membranes and vesicles that act as nucleation sites for crystals, promoting their formation. Damaged cells also facilitate crystal deposition and aggregation on the renal tubular cell surfaces, directly contributing to stone formation (Fasano and Khan, 2001; Khan, 2011). However, data from fruit flies fed a combination of NAC and NaOx showed that merely inhibiting oxidative stress does not completely treat kidney stones. Thus, while reducing oxidative stress supports stone prevention, it is not the primary mechanism for treating kidney stones.

Increasing urination is also a key mechanism in treating kidney stones as it reduces the concentration of calcium and oxalate in the urine and helps flush out small stones and fragments (Chen et al., 2020). Common diuretics used clinically for kidney stones treatment, such as furosemide and spironolactone, work by increasing urine production (Lyll et al., 2023). Haijinsha (*Spora Lygodii*) and Jinqiancao (*Lysimachia christinae*) are known to enhance ureter peristalsis, facilitate stone expulsion, and boost urine production, thereby lowering calcium

and oxalate concentrations, preventing stone formation (Gohel and Wong, 2006).

In this study, we identified that the water extracts of HH and its core compound HYA primarily treat kidney stones by affecting *in vitro* crystallization and reducing oxidative stress. DS and its core compound SAA exhibit similar therapeutic mechanisms. HQ and its core compound CG, contribute to the treatment of kidney stones through their impact on crystallization and oxidative stress. The alignment of mechanisms between the aqueous extracts and their principal core compounds confirms the identification of effective active compounds for treating kidney stones. In addition, we found that even single compounds can exert therapeutic effects through multiple distinct mechanisms. These mechanisms function independently but ultimately contribute synergistically to the overall therapeutic outcome. Moreover, the reduction in stone burden helps alleviate inflammation and oxidative stress, promotes the recovery of urine output, and establishes a positive feedback loop that further enhances the therapeutic efficacy.

To validate the effectiveness of the core compounds from these TCMs in mammals, we used a mouse model of kidney stones induced by intraperitoneal injection Gly. Our results indicate that these core components replicate their therapeutic effects in mammalian kidney stones, supporting the clinical relevance of our screening strategy and its potential for successful translation to clinical applications.

Compared to mouse models, our strategy using the fruit fly model to screen effective TCMs and identify their therapeutic mechanisms

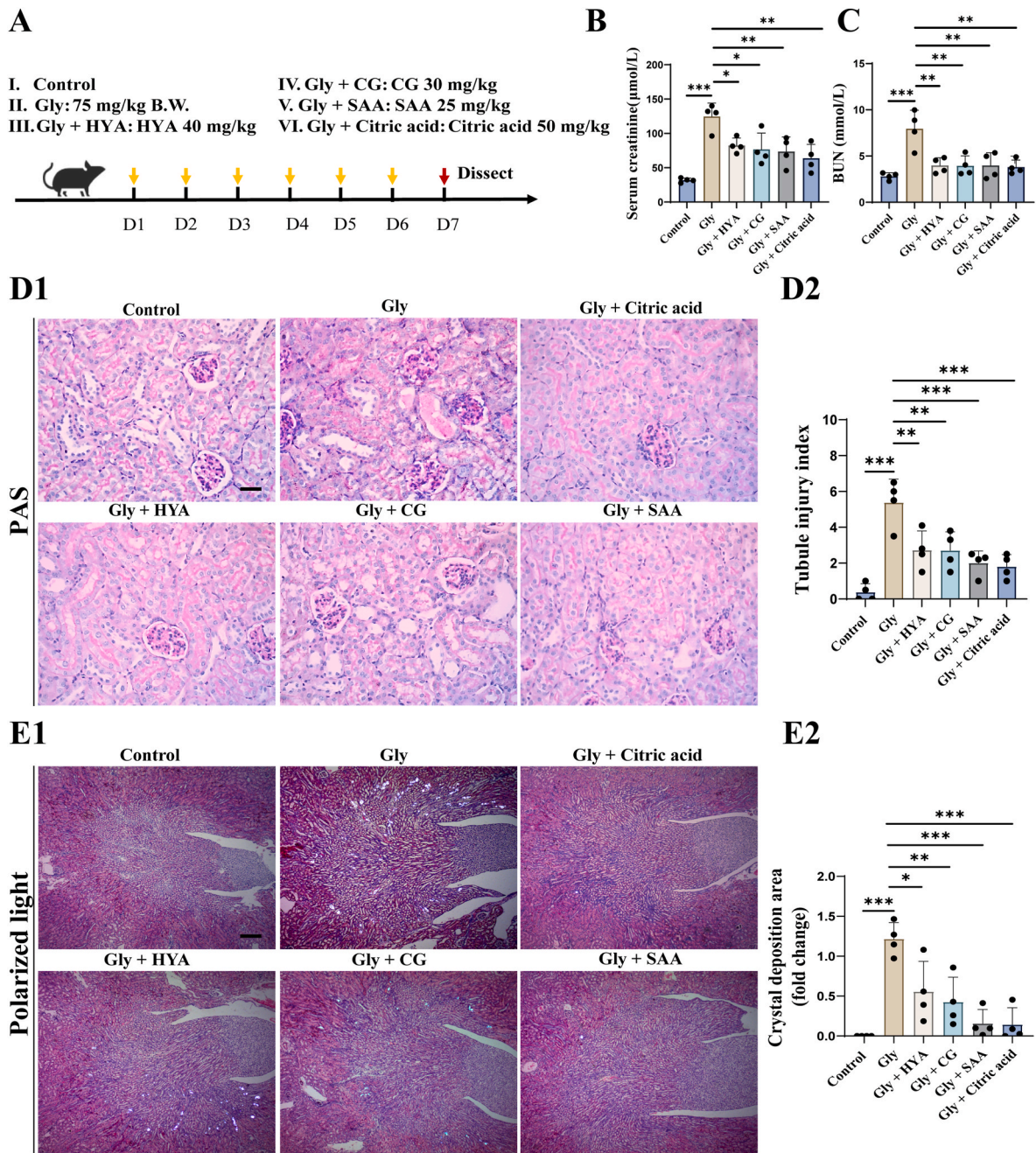


Fig. 8. HYA, SAA, and CG exhibit protective effects against kidney damage and inhibit renal calculus formation in mouse model. Schematic outlining the experimental design and group allocations (A). Representative the levels of Serum creatinine (B) and BUN (C) in the kidneys of mice. n=4. Representative images of renal CaOx crystal deposition via PAS staining of kidney tissue (D1) and polarized light microscopy (E1). Percentage histogram depicts varying degrees of scores of tubular injuries (D2) and CaOx crystals in mouse kidneys (E2). Scale bar: 100 μm, Data are represented as mean ± standard deviation (Mean ± SD), and analyzed by One-Way ANOVA (n/s, p>0.05; *, p<0.05; **, p<0.01; ***, p<0.001).

Table 1
Application of three herbs in the treatment of kidney stones.

Herbs	Botanical name	Voucher number	Key active compound	Prescriptions with corresponding herbs	Modern clinical research
Honghua	<i>Carthamus tinctorius</i> L.	2024YWY-2	Hydroxysafflower yellow A (HYA)	Xiaojian Paishi decoction	Jia and Zhang (2021) Chen et al. (2012)
Danshen	<i>Salvia miltiorrhiza</i> Bunge	2024YWY-3	Salvianic acid A (SAA)	Yiqi Paishi decoction	Sun et al. (2008)
Huangqi	<i>Astragalus membranaceus</i> (Fisch.) Bunge	2024YWY-4	Calycosin-7-O-β-D-glucoside (CG)	Huangqi Jinhu Paishi decoction	(Jiang, 2009; Zhao et al., 2018) (Wang and Song, 2010)

significantly reduces costs, accelerates research and development, and offers ethical advantages. The low maintenance cost of fruit fly allows for experiments with larger sample sizes, facilitating high-throughput screening for active compounds. Its rapid life cycle combined with powerful genetic tools enables efficient mechanistic studies. The active compounds and therapeutic pathways identified through this platform can be more precisely validated in mammalian models, greatly accelerating research progress while reducing both overall costs and the reliance on mammalian testing, which aligns with the 3R principles in animal research by minimizing ethical concerns. Overall, this strategy holds great promise for advancing the clinical and scientific use of herbal resources from diverse traditional medicine systems across various cultures and ethnic groups through the discovery of bioactive natural products.

Human kidney stone formation is a multifactorial and long-term process. Neither the mouse nor the fruit fly models in this work—both of which are typically based on hyperoxaluria—can fully replicate all aspects of human stone pathogenesis (Tzou et al., 2016). Nevertheless, the Gly induced mouse kidney stone model remains the most widely used and established model in current kidney stone research. When a compound shows therapeutic efficacy in both the fruit fly and mouse models, it suggests that the underlying mechanism is evolutionarily conserved, thereby increasing confidence in its druggability. In the future, additional fruit fly models can be further developed to reflect additional pathogenic factors—for instance, by knocking down *Agt* to elevate endogenous oxalate levels—thus expanding and refining the discovering active compounds for nephrolithiasis treatment from herbal medicines pipeline based on the fruit fly platform.

Fruit fly can also be used to establish a wide range of disease models, including diabetes (Li et al., 2023), inflammatory bowel disease (IBD) (Wen et al., 2024), cancer (Ferreira, 2024), and neurodegenerative disorders (Wu et al., 2023; Zhang et al., 2024), enabling the development of similar pipelines for active compound discovery. Therefore, this approach holds significant promise for advancing the clinical and scientific utilization of herbal resources by identifying active natural products for the treatment of various diseases.

CRediT authorship contribution statement

Yunuo Ren: Writing – original draft, Methodology, Data curation, Conceptualization. **Wen Zhang:** Methodology, Data curation, Conceptualization. **Youxin Li:** Methodology, Conceptualization. **Dekun Wang:** Methodology, Data curation, Conceptualization. **Weiwei Tang:** Methodology, Data curation, Conceptualization. **Lingfei Wang:** Methodology, Data curation, Conceptualization. **Xinghua Jin:** Methodology, Data curation, Conceptualization. **Donghai Xia:** Methodology, Conceptualization. **Wei Yu:** Methodology, Conceptualization. **Yaodong Miao:** Methodology, Conceptualization. **Yingjie Ju:** Methodology, Conceptualization. **Shushen Sun:** Methodology, Conceptualization. **Tianxiang Li:** Methodology, Conceptualization. **Shaoshan Du:** Writing – review & editing. **Zhimou Guo:** Writing – review & editing, Conceptualization. **Xiaoyue Tan:** Writing – review & editing, Conceptualization. **Zhiguang Yuchi:** Writing – review & editing, Conceptualization. **Yiwen Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.119846>.

Data availability

Data will be made available on request.

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